

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z2,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z2 ZL: $\frac{Z_1Z_2Z_Lg_m+Z_1Z_L}{Z_1Z_2g_m+Z_1+Z_2+Z_L}$

$$H(z) = \frac{Z_1Z_2Z_Lg_m + Z_1Z_L}{Z_1Z_2g_m + Z_1 + Z_2 + Z_L}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1s}, \ R_2, \ \infty, \ \infty, \ \infty, \ \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{R_2R_Lg_m + R_L}{C_1C_LR_2R_Ls^2 + R_2g_m + s(C_1R_2 + C_1R_L + C_LR_2R_Lg_m + C_LR_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_2}\sqrt{R_L}\sqrt{R_2g_m+1}}{C_1R_2+C_1R_L+C_LR_2R_Lg_m+C_LR_L}$
wo: $\frac{\sqrt{R_2g_m+1}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_2}\sqrt{R_L}}$
bandwidth: $\frac{C_1R_2+C_1R_L+C_LR_2R_Lg_m+C_LR_L}{C_1C_LR_2R_L}$
K-LP: R_L
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \ R_2, \ \infty, \ \infty, \ \infty, \ \frac{1}{C_Ls} \right)$

$$H(s) = \frac{R_1R_2g_m + R_1}{C_1C_LR_1R_2s^2 + s(C_1R_1 + C_LR_1R_2g_m + C_LR_1 + C_LR_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_2}}{C_1R_1+C_LR_1R_2g_m+C_LR_1+C_LR_2}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_2}}$
bandwidth: $\frac{C_1R_1+C_LR_1R_2g_m+C_LR_1+C_LR_2}{C_1C_LR_1R_2}$
K-LP: $R_1R_2g_m + R_1$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_2 R_L g_m + R_1 R_L}{C_1 C_L R_1 R_2 R_L s^2 + R_1 R_2 g_m + R_1 + R_2 + R_L + s(C_1 R_1 R_2 + C_1 R_1 R_L + C_L R_1 R_2 R_L g_m + C_L R_1 R_L + C_L R_2 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_2} \sqrt{R_L} \sqrt{R_1 R_2 g_m + R_1 + R_2 + R_L}}{C_1 R_1 R_2 + C_1 R_1 R_L + C_L R_1 R_2 R_L g_m + C_L R_1 R_L + C_L R_2 R_L}$
 wo: $\frac{\sqrt{R_1 R_2 g_m + R_1 + R_2 + R_L}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_2} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 R_1 R_2 + C_1 R_1 R_L + C_L R_1 R_2 R_L g_m + C_L R_1 R_L + C_L R_2 R_L}{C_1 C_L R_1 R_2 R_L}$
 K-LP: $\frac{R_1 R_2 R_L g_m + R_1 R_L}{R_1 R_2 g_m + R_1 + R_2 + R_L}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_2 R_1 R_L s + R_1 R_L g_m}{C_2 C_L R_1 R_L s^2 + R_1 g_m + s(C_2 R_1 + C_2 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_2} \sqrt{C_L} \sqrt{R_1} \sqrt{R_L} \sqrt{R_1 g_m + 1}}{C_2 R_1 + C_2 R_L + C_L R_1 R_L g_m + C_L R_L}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_2} \sqrt{C_L} \sqrt{R_1} \sqrt{R_L}}$
 bandwidth: $\frac{C_2 R_1 + C_2 R_L + C_L R_1 R_L g_m + C_L R_L}{C_2 C_L R_1 R_L}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_L}{C_2 R_1 + C_2 R_L + C_L R_1 R_L g_m + C_L R_L}$
 Qz: None
 Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_2 R_1 R_2 s + R_1 R_2 g_m + R_1}{C_2 C_L R_1 R_2 s^2 + s(C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_2} \sqrt{C_L} \sqrt{R_1} \sqrt{R_2}}{C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2}$
 wo: $\frac{1}{\sqrt{C_2} \sqrt{C_L} \sqrt{R_1} \sqrt{R_2}}$
 bandwidth: $\frac{C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2}{C_2 C_L R_1 R_2}$
 K-LP: $R_1 R_2 g_m + R_1$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_2}{C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2}$
 Qz: None
 Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_2 R_1 R_2 R_L s + R_1 R_2 R_L g_m + R_1 R_L}{C_2 C_L R_1 R_2 R_L s^2 + R_1 R_2 g_m + R_1 + R_2 + R_L + s(C_2 R_1 R_2 + C_2 R_2 R_L + C_L R_1 R_2 R_L g_m + C_L R_1 R_L + C_L R_2 R_L)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_2}\sqrt{C_L}\sqrt{R_1}\sqrt{R_2}\sqrt{R_L}\sqrt{R_1R_2g_m+R_1+R_2+R_L}}{C_2R_1R_2+C_2R_2R_L+C_LR_1R_2R_Lg_m+C_LR_1R_L+C_LR_2R_L} \\
\text{wo: } & \frac{\sqrt{R_1R_2g_m+R_1+R_2+R_L}}{\sqrt{C_2}\sqrt{C_L}\sqrt{R_1}\sqrt{R_2}\sqrt{R_L}} \\
\text{bandwidth: } & \frac{C_2R_1R_2+C_2R_2R_L+C_LR_1R_2R_Lg_m+C_LR_1R_L+C_LR_2R_L}{C_2C_LR_1R_2R_L} \\
\text{K-LP: } & \frac{R_1R_2R_Lg_m+R_1R_L}{R_1R_2g_m+R_1+R_2+R_L} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_2R_1R_2R_L}{C_2R_1R_2+C_2R_2R_L+C_LR_1R_2R_Lg_m+C_LR_1R_L+C_LR_2R_L} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(R_1, R_2 + \frac{1}{C_2s}, \infty, \infty, \infty, \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{R_1R_Lg_m + s(C_2R_1R_2R_Lg_m + C_2R_1R_L)}{R_1g_m + s^2(C_2C_LR_1R_2R_Lg_m + C_2C_LR_1R_L + C_2C_LR_2R_L) + s(C_2R_1R_2g_m + C_2R_1 + C_2R_2 + C_2R_L + C_LR_1R_Lg_m + C_LR_L) + 1}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_2}\sqrt{C_L}R_1R_2\sqrt{R_L}g_m\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}} + \sqrt{C_2}\sqrt{C_L}R_1\sqrt{R_L}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}} + \sqrt{C_2}\sqrt{C_L}R_2\sqrt{R_L}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}}}{C_2R_1R_2g_m + C_2R_1 + C_2R_2 + C_2R_L + C_LR_1R_Lg_m + C_LR_L} \\
\text{wo: } & \sqrt{\frac{R_1g_m+1}{C_2C_LR_1R_2R_Lg_m + C_2C_LR_1R_L + C_2C_LR_2R_L}} \\
\text{bandwidth: } & \frac{\sqrt{\frac{R_1g_m+1}{C_2C_LR_1R_2R_Lg_m + C_2C_LR_1R_L + C_2C_LR_2R_L}}(C_2R_1R_2g_m + C_2R_1 + C_2R_2 + C_2R_L + C_LR_1R_Lg_m + C_LR_L)}{\sqrt{C_2}\sqrt{C_L}R_1R_2\sqrt{R_L}g_m\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}} + \sqrt{C_2}\sqrt{C_L}R_1\sqrt{R_L}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}} + \sqrt{C_2}\sqrt{C_L}R_2\sqrt{R_L}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}}} \\
\text{K-LP: } & \frac{R_1R_Lg_m}{R_1g_m+1} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_2R_1R_2R_Lg_m + C_2R_1R_L}{C_2R_1R_2g_m + C_2R_1 + C_2R_2 + C_2R_L + C_LR_1R_Lg_m + C_LR_L} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{1}{C_1s}, \frac{1}{C_2s}, \infty, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_2R_Ls + R_Lg_m}{C_1C_2R_Ls^2 + g_m + s(C_1 + C_2)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_1}\sqrt{C_2}\sqrt{R_L}\sqrt{g_m}}{C_1+C_2} \\
\text{wo: } & \frac{\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_L}} \\
\text{bandwidth: } & \frac{C_1+C_2}{C_1C_2R_L} \\
\text{K-LP: } & R_L \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_2R_L}{C_1+C_2} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{1}{C_1s}, \frac{1}{C_2s}, \infty, \infty, \infty, \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{C_2R_Ls + R_Lg_m}{g_m + s^2(C_1C_2R_L + C_1C_LR_L + C_2C_LR_L) + s(C_1 + C_2 + C_LR_Lg_m)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{R_L}\sqrt{g_m}\sqrt{C_1C_2+C_1C_L+C_2C_L}}{C_1+C_2+C_LR_Lg_m} \\
\text{wo: } & \frac{\sqrt{g_m}}{\sqrt{C_1C_2R_L+C_1C_LR_L+C_2C_LR_L}} \\
\text{bandwidth: } & \frac{C_1+C_2+C_LR_Lg_m}{\sqrt{R_L}\sqrt{C_1C_2+C_1C_L+C_2C_L}\sqrt{C_1C_2R_L+C_1C_LR_L+C_2C_LR_L}} \\
\text{K-LP: } & R_L \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_2R_L}{C_1+C_2+C_LR_Lg_m} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_2 R_2 R_L s + R_2 R_L g_m + R_L}{C_1 C_2 R_2 R_L s^2 + R_2 g_m + s(C_1 R_2 + C_1 R_L + C_2 R_2) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_2} \sqrt{R_2} \sqrt{R_L} \sqrt{R_2 g_m + 1}}{C_1 R_2 + C_1 R_L + C_2 R_2} \\ \text{wo: } & \frac{\sqrt{R_2 g_m + 1}}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_2} \sqrt{R_L}} \\ \text{bandwidth: } & \frac{C_1 R_2 + C_1 R_L + C_2 R_2}{C_1 C_2 R_2 R_L} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_2 R_2 R_L}{C_1 R_2 + C_1 R_L + C_2 R_2} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_2 R_2 R_L s + R_2 R_L g_m + R_L}{R_2 g_m + s^2(C_1 C_2 R_2 R_L + C_1 C_L R_2 R_L + C_2 C_L R_2 R_L) + s(C_1 R_2 + C_1 R_L + C_2 R_2 + C_L R_2 R_L g_m + C_L R_L) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{R_2} \sqrt{R_L} \sqrt{R_2 g_m + 1} \sqrt{C_1 C_2 + C_1 C_L + C_2 C_L}}{C_1 R_2 + C_1 R_L + C_2 R_2 + C_L R_2 R_L g_m + C_L R_L} \\ \text{wo: } & \frac{\sqrt{R_2 g_m + 1}}{\sqrt{C_1 C_2 R_2 R_L + C_1 C_L R_2 R_L + C_2 C_L R_2 R_L}} \\ \text{bandwidth: } & \frac{C_1 R_2 + C_1 R_L + C_2 R_2 + C_L R_2 R_L g_m + C_L R_L}{\sqrt{R_2} \sqrt{R_L} \sqrt{C_1 C_2 + C_1 C_L + C_2 C_L} \sqrt{C_1 C_2 R_2 R_L + C_1 C_L R_2 R_L + C_2 C_L R_2 R_L}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_2 R_2 R_L}{C_1 R_2 + C_1 R_L + C_2 R_2 + C_L R_2 R_L g_m + C_L R_L} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, R_L \right)$

$$H(s) = \frac{R_L g_m + s(C_2 R_2 R_L g_m + C_2 R_L)}{g_m + s^2(C_1 C_2 R_2 + C_1 C_2 R_L) + s(C_1 + C_2 R_2 g_m + C_2)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_2} \sqrt{g_m} \sqrt{R_2 + R_L}}{C_1 + C_2 R_2 g_m + C_2} \\ \text{wo: } & \frac{\sqrt{g_m}}{\sqrt{C_1 C_2 R_2 + C_1 C_2 R_L}} \\ \text{bandwidth: } & \frac{C_1 + C_2 R_2 g_m + C_2}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_2 + R_L} \sqrt{C_1 C_2 R_2 + C_1 C_2 R_L}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_2 R_2 R_L g_m + C_2 R_L}{C_1 + C_2 R_2 g_m + C_2} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s(C_L R_1 R_2 R_L g_m + C_L R_1 R_L)}{s^2(C_1 C_L R_1 R_2 + C_1 C_L R_1 R_L) + s(C_1 R_1 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_2 + R_L}}{C_1 R_1 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L} \\ \text{wo: } & \frac{1}{\sqrt{C_1 C_L R_1 R_2 + C_1 C_L R_1 R_L}} \\ \text{bandwidth: } & \frac{C_1 R_1 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_2 + R_L} \sqrt{C_1 C_L R_1 R_2 + C_1 C_L R_1 R_L}} \end{aligned}$$

K-LP: $R_1 R_2 g_m + R_1$
K-HP: 0
K-BP: $\frac{C_L R_1 R_2 R_L g_m + C_L R_1 R_L}{C_1 R_1 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L}$
Qz: None
Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{1}{C_2 s}, \infty, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_2 R_1 R_L s + R_1 R_L g_m}{C_1 C_2 R_1 R_L s^2 + R_1 g_m + s(C_1 R_1 + C_2 R_1 + C_2 R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_2 R_1 + C_2 R_L}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_L}}$
bandwidth: $\frac{C_1 R_1 + C_2 R_1 + C_2 R_L}{C_1 C_2 R_1 R_L}$
K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_L}{C_1 R_1 + C_2 R_1 + C_2 R_L}$
Qz: None
Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{1}{C_2 s}, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_2 R_1 R_L s + R_1 R_L g_m}{R_1 g_m + s^2(C_1 C_2 R_1 R_L + C_1 C_L R_1 R_L + C_2 C_L R_1 R_L) + s(C_1 R_1 + C_2 R_1 + C_2 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_L} \sqrt{R_1 g_m + 1} \sqrt{C_1 C_2 + C_1 C_L + C_2 C_L}}{C_1 R_1 + C_2 R_1 + C_2 R_L + C_L R_1 R_L g_m + C_L R_L}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1 C_2 R_1 R_L + C_1 C_L R_1 R_L + C_2 C_L R_1 R_L}}$
bandwidth: $\frac{C_1 R_1 + C_2 R_1 + C_2 R_L + C_L R_1 R_L g_m + C_L R_L}{\sqrt{R_1} \sqrt{R_L} \sqrt{C_1 C_2 + C_1 C_L + C_2 C_L} \sqrt{C_1 C_2 R_1 R_L + C_1 C_L R_1 R_L + C_2 C_L R_1 R_L}}$
K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_L}{C_1 R_1 + C_2 R_1 + C_2 R_L + C_L R_1 R_L g_m + C_L R_L}$
Qz: None
Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_2 R_1 R_2 R_L s + R_1 R_2 R_L g_m + R_1 R_L}{C_1 C_2 R_1 R_2 R_L s^2 + R_1 R_2 g_m + R_1 + R_2 + R_L + s(C_1 R_1 R_2 + C_1 R_1 R_L + C_2 R_1 R_2 + C_2 R_2 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2} \sqrt{R_L} \sqrt{R_1 R_2 g_m + R_1 + R_2 + R_L}}{C_1 R_1 R_2 + C_1 R_1 R_L + C_2 R_1 R_2 + C_2 R_2 R_L}$
wo: $\frac{\sqrt{R_1 R_2 g_m + R_1 + R_2 + R_L}}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2} \sqrt{R_L}}$
bandwidth: $\frac{C_1 R_1 R_2 + C_1 R_1 R_L + C_2 R_1 R_2 + C_2 R_2 R_L}{C_1 C_2 R_1 R_2 R_L}$
K-LP: $\frac{R_1 R_2 R_L g_m + R_1 R_L}{R_1 R_2 g_m + R_1 + R_2 + R_L}$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_2 R_L}{C_1 R_1 R_2 + C_1 R_1 R_L + C_2 R_1 R_2 + C_2 R_2 R_L}$
Qz: None
Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_2 R_1 R_2 s + R_1 R_2 g_m + R_1}{s^2 (C_1 C_2 R_1 R_2 + C_1 C_L R_1 R_2 + C_2 C_L R_1 R_2) + s (C_1 R_1 + C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_2} \sqrt{C_1 C_2 + C_1 C_L + C_2 C_L}}{C_1 R_1 + C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2}$
 wo: $\frac{1}{\sqrt{C_1 C_2 R_1 R_2 + C_1 C_L R_1 R_2 + C_2 C_L R_1 R_2}}$
 bandwidth: $\frac{C_1 R_1 + C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2}{\sqrt{R_1} \sqrt{R_2} \sqrt{C_1 C_2 + C_1 C_L + C_2 C_L} \sqrt{C_1 C_2 R_1 R_2 + C_1 C_L R_1 R_2 + C_2 C_L R_1 R_2}}$
 K-LP: $R_1 R_2 g_m + R_1$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_2}{C_1 R_1 + C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2}$
 Qz: None
 Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_2 R_1 R_2 R_L s + R_1 R_2 R_L g_m + R_1 R_L}{R_1 R_2 g_m + R_1 + R_2 + R_L + s^2 (C_1 C_2 R_1 R_2 R_L + C_1 C_L R_1 R_2 R_L + C_2 C_L R_1 R_2 R_L) + s (C_1 R_1 R_2 + C_1 R_1 R_L + C_2 R_1 R_2 + C_2 R_2 R_L + C_L R_1 R_2 R_L g_m + C_L R_1 R_L + C_L R_2 R_L)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_2} \sqrt{R_L} \sqrt{C_1 C_2 + C_1 C_L + C_2 C_L} \sqrt{R_1 R_2 g_m + R_1 + R_2 + R_L}}{C_1 R_1 R_2 + C_1 R_1 R_L + C_2 R_1 R_2 + C_2 R_2 R_L + C_L R_1 R_2 R_L g_m + C_L R_1 R_L + C_L R_2 R_L}$
 wo: $\frac{\sqrt{R_1 R_2 g_m + R_1 + R_2 + R_L}}{\sqrt{C_1 C_2 R_1 R_2 R_L + C_1 C_L R_1 R_2 R_L + C_2 C_L R_1 R_2 R_L}}$
 bandwidth: $\frac{C_1 R_1 R_2 + C_1 R_1 R_L + C_2 R_1 R_2 + C_2 R_2 R_L + C_L R_1 R_2 R_L g_m + C_L R_1 R_L + C_L R_2 R_L}{\sqrt{R_1} \sqrt{R_2} \sqrt{R_L} \sqrt{C_1 C_2 + C_1 C_L + C_2 C_L} \sqrt{C_1 C_2 R_1 R_2 R_L + C_1 C_L R_1 R_2 R_L + C_2 C_L R_1 R_2 R_L}}$
 K-LP: $\frac{R_1 R_2 R_L g_m + R_1 R_L}{R_1 R_2 g_m + R_1 + R_2 + R_L}$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_2 R_L}{C_1 R_1 R_2 + C_1 R_1 R_L + C_2 R_1 R_2 + C_2 R_2 R_L + C_L R_1 R_2 R_L g_m + C_L R_1 R_L + C_L R_2 R_L}$
 Qz: None
 Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_L g_m + s (C_2 R_1 R_2 R_L g_m + C_2 R_1 R_L)}{R_1 g_m + s^2 (C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_L) + s (C_1 R_1 + C_2 R_1 R_2 g_m + C_2 R_1 + C_2 R_2 + C_2 R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2 + R_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_2 R_1 R_2 g_m + C_2 R_1 + C_2 R_2 + C_2 R_L}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_L}}$
 bandwidth: $\frac{C_1 R_1 + C_2 R_1 R_2 g_m + C_2 R_1 + C_2 R_2 + C_2 R_L}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2 + R_L} \sqrt{C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_L}}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_2 R_L g_m + C_2 R_1 R_L}{C_1 R_1 + C_2 R_1 R_2 g_m + C_2 R_1 + C_2 R_2 + C_2 R_L}$
 Qz: None
 Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_2 R_L g_m + R_L + s (C_1 R_1 R_2 R_L g_m + C_1 R_1 R_L)}{R_2 g_m + s^2 (C_1 C_L R_1 R_2 R_L g_m + C_1 C_L R_1 R_L + C_1 C_L R_2 R_L) + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L + C_L R_2 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} R_1 R_2 \sqrt{R_L} g_m \sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}} + \sqrt{C_1} \sqrt{C_L} R_1 \sqrt{R_L} \sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}} + \sqrt{C_1} \sqrt{C_L} R_2 \sqrt{R_L} \sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}}}{C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L + C_L R_2 R_L g_m + C_L R_L}$
 wo: $\sqrt{\frac{R_2 g_m + 1}{C_1 C_L R_1 R_2 R_L g_m + C_1 C_L R_1 R_L + C_1 C_L R_2 R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_2 g_m + 1}{C_1 C_L R_1 R_2 R_L g_m + C_1 C_L R_1 R_L + C_1 C_L R_2 R_L}} (C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L + C_L R_2 R_L g_m + C_L R_L)}{\sqrt{C_1} \sqrt{C_L} R_1 R_2 \sqrt{R_L g_m} \sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}} + \sqrt{C_1} \sqrt{C_L} R_1 \sqrt{R_L} \sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}} + \sqrt{C_1} \sqrt{C_L} R_2 \sqrt{R_L} \sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}}}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_1 R_1 R_2 R_L g_m + C_1 R_1 R_L}{C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L + C_L R_2 R_L g_m + C_L R_L}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, R_2, \infty, \infty, \infty, R_L)$

$$H(s) = \frac{R_1 R_2 R_L g_m + R_1 R_L}{R_1 R_2 g_m + R_1 + R_2 + R_L}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, R_2, \infty, \infty, \infty, \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_2 g_m + R_1}{s(C_L R_1 R_2 g_m + C_L R_1 + C_L R_2) + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, R_2, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_1 R_2 R_L g_m + R_1 R_L}{R_1 R_2 g_m + R_1 + R_2 + R_L + s(C_L R_1 R_2 R_L g_m + C_L R_1 R_L + C_L R_2 R_L)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, R_2, \infty, \infty, \infty, R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s(C_L R_1 R_2 R_L g_m + C_L R_1 R_L)}{s(C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \infty, \infty, R_L\right)$

$$H(s) = \frac{C_2 R_1 R_L s + R_1 R_L g_m}{R_1 g_m + s(C_2 R_1 + C_2 R_L) + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \infty, \infty, \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 R_1 s + R_1 g_m}{C_2 C_L R_1 s^2 + s(C_2 + C_L R_1 g_m + C_L)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \infty, \infty, R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L R_1 R_L s^2 + R_1 g_m + s(C_2 R_1 + C_L R_1 R_L g_m)}{s^2(C_2 C_L R_1 + C_2 C_L R_L) + s(C_2 + C_L R_1 g_m + C_L)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, R_L\right)$

$$H(s) = \frac{C_2 R_1 R_2 R_L s + R_1 R_2 R_L g_m + R_1 R_L}{R_1 R_2 g_m + R_1 + R_2 + R_L + s(C_2 R_1 R_2 + C_2 R_2 R_L)}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(R_1, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_1 R_L g_m + s (C_2 R_1 R_2 R_L g_m + C_2 R_1 R_L)}{R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1 + C_2 R_2 + C_2 R_L) + 1}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(R_1, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{s^2 (C_2 C_L R_1 R_2 g_m + C_2 C_L R_1 + C_2 C_L R_2) + s (C_2 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_2 C_L R_1 R_2 R_L g_m + C_2 C_L R_1 R_L) + s (C_2 R_1 R_2 g_m + C_2 R_1 + C_L R_1 R_L g_m)}{s^2 (C_2 C_L R_1 R_2 g_m + C_2 C_L R_1 + C_2 C_L R_2 + C_2 C_L R_L) + s (C_2 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad \infty, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_2 R_L g_m + R_L}{R_2 g_m + s (C_1 R_2 + C_1 R_L) + 1}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_2 g_m + 1}{C_1 C_L R_2 s^2 + s (C_1 + C_L R_2 g_m + C_L)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_2 g_m + s (C_L R_2 R_L g_m + C_L R_L) + 1}{s^2 (C_1 C_L R_2 + C_1 C_L R_L) + s (C_1 + C_L R_2 g_m + C_L)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_2 s + g_m}{C_L g_m s + s^2 (C_1 C_2 + C_1 C_L + C_2 C_L)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_2 C_L R_L s^2 + g_m + s (C_2 + C_L R_L g_m)}{C_1 C_2 C_L R_L s^3 + C_L g_m s + s^2 (C_1 C_2 + C_1 C_L + C_2 C_L)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_2 R_2 s + R_2 g_m + 1}{s^2 (C_1 C_2 R_2 + C_1 C_L R_2 + C_2 C_L R_2) + s (C_1 + C_L R_2 g_m + C_L)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_2 C_L R_2 R_L s^2 + R_2 g_m + s (C_2 R_2 + C_L R_2 R_L g_m + C_L R_L) + 1}{C_1 C_2 C_L R_2 R_L s^3 + s^2 (C_1 C_2 R_2 + C_1 C_L R_2 + C_1 C_L R_L + C_2 C_L R_2) + s (C_1 + C_L R_2 g_m + C_L)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s (C_2 R_2 g_m + C_2)}{C_1 C_2 C_L R_2 s^3 + C_L g_m s + s^2 (C_1 C_2 + C_1 C_L + C_2 C_L R_2 g_m + C_2 C_L)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L g_m + s (C_2 R_2 R_L g_m + C_2 R_L)}{C_1 C_2 C_L R_2 R_L s^3 + g_m + s^2 (C_1 C_2 R_2 + C_1 C_2 R_L + C_1 C_L R_L + C_2 C_L R_2 R_L g_m + C_2 C_L R_L) + s (C_1 + C_2 R_2 g_m + C_2 + C_L R_L g_m)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s^2 (C_2 C_L R_2 R_L g_m + C_2 C_L R_L) + s (C_2 R_2 g_m + C_2 + C_L R_L g_m)}{C_L g_m s + s^3 (C_1 C_2 C_L R_2 + C_1 C_2 C_L R_L) + s^2 (C_1 C_2 + C_1 C_L + C_2 C_L R_2 g_m + C_2 C_L)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2, \quad \infty, \quad \infty, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_1 R_2 R_L g_m + R_1 R_L}{R_1 R_2 g_m + R_1 + R_2 + R_L + s (C_1 R_1 R_2 + C_1 R_1 R_L)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_2 R_1 s + R_1 g_m}{s^2 (C_1 C_2 R_1 + C_1 C_L R_1 + C_2 C_L R_1) + s (C_2 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_2 C_L R_1 R_L s^2 + R_1 g_m + s (C_2 R_1 + C_L R_1 R_L g_m)}{C_1 C_2 C_L R_1 R_L s^3 + s^2 (C_1 C_2 R_1 + C_1 C_L R_1 + C_2 C_L R_1 + C_2 C_L R_L) + s (C_2 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_2 C_L R_1 R_2 R_L s^2 + R_1 R_2 g_m + R_1 + s (C_2 R_1 R_2 + C_L R_1 R_2 R_L g_m + C_L R_1 R_L)}{C_1 C_2 C_L R_1 R_2 R_L s^3 + s^2 (C_1 C_2 R_1 R_2 + C_1 C_L R_1 R_2 + C_1 C_L R_1 R_L + C_2 C_L R_1 R_2 + C_2 C_L R_2 R_L) + s (C_1 R_1 + C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L) + 1}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{C_1 C_2 C_L R_1 R_2 s^3 + s^2 (C_1 C_2 R_1 + C_1 C_L R_1 + C_2 C_L R_1 R_2 g_m + C_2 C_L R_1 + C_2 C_L R_2) + s (C_2 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_1 R_L g_m + s (C_2 R_1 R_2 R_L g_m + C_2 R_1 R_L)}{C_1 C_2 C_L R_1 R_2 R_L s^3 + R_1 g_m + s^2 (C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_L + C_1 C_L R_1 R_L + C_2 C_L R_1 R_2 R_L g_m + C_2 C_L R_1 R_L + C_2 C_L R_2 R_L) + s (C_1 R_1 + C_2 R_1 R_2 g_m + C_2 R_1 + C_2 R_2 + C_2 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_2 C_L R_1 R_2 R_L g_m + C_2 C_L R_1 R_L) + s (C_2 R_1 R_2 g_m + C_2 R_1 + C_L R_1 R_L g_m)}{s^3 (C_1 C_2 C_L R_1 R_2 + C_1 C_2 C_L R_1 R_L) + s^2 (C_1 C_2 R_1 + C_1 C_L R_1 + C_2 C_L R_1 R_2 g_m + C_2 C_L R_1 + C_2 C_L R_2 + C_2 C_L R_L) + s (C_2 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad \infty, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_2 R_L g_m + R_L + s(C_1 R_1 R_2 R_L g_m + C_1 R_1 R_L)}{R_2 g_m + s(C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L) + 1}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_2 g_m + s(C_1 R_1 R_2 g_m + C_1 R_1) + 1}{s^2(C_1 C_L R_1 R_2 g_m + C_1 C_L R_1 + C_1 C_L R_2) + s(C_1 + C_L R_2 g_m + C_L)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_2 g_m + s^2(C_1 C_L R_1 R_2 R_L g_m + C_1 C_L R_1 R_L) + s(C_1 R_1 R_2 g_m + C_1 R_1 + C_L R_2 R_L g_m + C_L R_L) + 1}{s^2(C_1 C_L R_1 R_2 g_m + C_1 C_L R_1 + C_1 C_L R_2 + C_1 C_L R_L) + s(C_1 + C_L R_2 g_m + C_L)}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_2 R_1 s^2 + g_m + s(C_1 R_1 g_m + C_2)}{C_1 C_2 C_L R_1 s^3 + C_L g_m s + s^2(C_1 C_2 + C_1 C_L R_1 g_m + C_1 C_L + C_2 C_L)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_1 C_2 R_1 R_L s^2 + R_L g_m + s(C_1 R_1 R_L g_m + C_2 R_L)}{C_1 C_2 C_L R_1 R_L s^3 + g_m + s^2(C_1 C_2 R_1 + C_1 C_2 R_L + C_1 C_L R_1 R_L g_m + C_1 C_L R_L + C_2 C_L R_L) + s(C_1 R_1 g_m + C_1 + C_2 + C_L R_L g_m)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_2 C_L R_1 R_L s^3 + g_m + s^2(C_1 C_2 R_1 + C_1 C_L R_1 R_L g_m + C_2 C_L R_L) + s(C_1 R_1 g_m + C_2 + C_L R_L g_m)}{C_L g_m s + s^3(C_1 C_2 C_L R_1 + C_1 C_2 C_L R_L) + s^2(C_1 C_2 + C_1 C_L R_1 g_m + C_1 C_L + C_2 C_L)}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_2 R_1 R_2 s^2 + R_2 g_m + s(C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2) + 1}{C_1 C_2 C_L R_1 R_2 s^3 + s^2(C_1 C_2 R_2 + C_1 C_L R_1 R_2 g_m + C_1 C_L R_1 + C_1 C_L R_2 + C_2 C_L R_2) + s(C_1 + C_L R_2 g_m + C_L)}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_1 C_2 R_1 R_2 R_L s^2 + R_2 R_L g_m + R_L + s(C_1 R_1 R_2 R_L g_m + C_1 R_1 R_L + C_2 R_2 R_L)}{C_1 C_2 C_L R_1 R_2 R_L s^3 + R_2 g_m + s^2(C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_L + C_1 C_L R_1 R_2 R_L g_m + C_1 C_L R_1 R_L + C_1 C_L R_2 R_L + C_2 C_L R_2 R_L) + s(C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L + C_2 R_2 + C_L R_2 R_L g_m + C_L R_L) + 1}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_2 C_L R_1 R_2 R_L s^3 + R_2 g_m + s^2(C_1 C_2 R_1 R_2 + C_1 C_L R_1 R_2 R_L g_m + C_1 C_L R_1 R_L + C_2 C_L R_2 R_L) + s(C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2 + C_L R_2 R_L g_m + C_L R_L) + 1}{s^3(C_1 C_2 C_L R_1 R_2 + C_1 C_2 C_L R_2 R_L) + s^2(C_1 C_2 R_2 + C_1 C_L R_1 R_2 g_m + C_1 C_L R_1 + C_1 C_L R_2 + C_1 C_L R_L + C_2 C_L R_2) + s(C_1 + C_L R_2 g_m + C_L)}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s^2(C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1) + s(C_1 R_1 g_m + C_2 R_2 g_m + C_2)}{C_L g_m s + s^3(C_1 C_2 C_L R_1 R_2 g_m + C_1 C_2 C_L R_1 + C_1 C_2 C_L R_2) + s^2(C_1 C_2 + C_1 C_L R_1 g_m + C_1 C_L + C_2 C_L R_2 g_m + C_2 C_L)}$$

9.39 X-INVALID-ORDER-39 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m + s^2 (C_1 C_2 R_1 R_2 R_L g_m + C_1 C_2 R_1 R_L) + s (C_1 R_1 R_L g_m + C_2 R_2 R_L g_m + C_2 R_L)}{g_m + s^3 (C_1 C_2 C_L R_1 R_2 R_L g_m + C_1 C_2 C_L R_1 R_L + C_1 C_2 C_L R_2 R_L) + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_1 C_2 R_2 + C_1 C_2 R_L + C_1 C_L R_1 R_L g_m + C_1 C_L R_L + C_2 C_L R_2 R_L g_m + C_2 C_L R_L) + s (C_1 R_1 g_m + C_1 + C_2 R_2 g_m + C_2 + C_L R_L g_m)}$$

9.40 X-INVALID-ORDER-40 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m + s^3 (C_1 C_2 C_L R_1 R_2 R_L g_m + C_1 C_2 C_L R_1 R_L) + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_1 C_L R_1 R_L g_m + C_2 C_L R_2 R_L g_m + C_2 C_L R_L) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2 + C_L R_L g_m)}{C_L g_m s + s^3 (C_1 C_2 C_L R_1 R_2 g_m + C_1 C_2 C_L R_1 + C_1 C_2 C_L R_2 + C_1 C_2 C_L R_L) + s^2 (C_1 C_2 + C_1 C_L R_1 g_m + C_1 C_L + C_2 C_L R_2 g_m + C_2 C_L)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_2 C_L R_1 R_2 R_L s^2 + R_1 R_2 g_m + R_1 + s (C_2 R_1 R_2 + C_L R_1 R_2 R_L g_m + C_L R_1 R_L)}{s^2 (C_2 C_L R_1 R_2 + C_2 C_L R_2 R_L) + s (C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_2} \sqrt{C_L} \sqrt{R_2} \sqrt{R_1 + R_L}}{C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L}$
 wo: $\frac{1}{\sqrt{C_2 C_L R_1 R_2 + C_2 C_L R_2 R_L}}$
 bandwidth: $\frac{C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L}{\sqrt{C_2} \sqrt{C_L} \sqrt{R_2} \sqrt{R_1 + R_L} \sqrt{C_2 C_L R_1 R_2 + C_2 C_L R_2 R_L}}$
 K-LP: $R_1 R_2 g_m + R_1$
 K-HP: $\frac{R_1 R_L}{R_1 + R_L}$
 K-BP: $\frac{C_2 R_1 R_2 + C_L R_1 R_2 R_L g_m + C_L R_1 R_L}{C_2 R_2 + C_L R_1 R_2 g_m + C_L R_1 + C_L R_2 + C_L R_L}$
 Qz: None
 Wz: $\frac{\sqrt{R_2 g_m + 1}}{\sqrt{C_2} \sqrt{C_L} \sqrt{R_2} \sqrt{R_L}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_1 C_2 R_1 R_L s^2 + R_L g_m + s (C_1 R_1 R_L g_m + C_2 R_L)}{g_m + s^2 (C_1 C_2 R_1 + C_1 C_2 R_L) + s (C_1 R_1 g_m + C_1 + C_2)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_2} \sqrt{g_m} \sqrt{R_1 + R_L}}{C_1 R_1 g_m + C_1 + C_2}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1 C_2 R_1 + C_1 C_2 R_L}}$
 bandwidth: $\frac{C_1 R_1 g_m + C_1 + C_2}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1 + R_L} \sqrt{C_1 C_2 R_1 + C_1 C_2 R_L}}$
 K-LP: R_L
 K-HP: $\frac{R_1 R_L}{R_1 + R_L}$
 K-BP: $\frac{C_1 R_1 R_L g_m + C_2 R_L}{C_1 R_1 g_m + C_1 + C_2}$
 Qz: None
 Wz: $\frac{\sqrt{g_m}}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_1 C_2 R_1 R_2 R_L s^2 + R_2 R_L g_m + R_L + s (C_1 R_1 R_2 R_L g_m + C_1 R_1 R_L + C_2 R_2 R_L)}{R_2 g_m + s^2 (C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_L) + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L + C_2 R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_2} \sqrt{R_2} \sqrt{R_1 + R_L} \sqrt{R_2 g_m + 1}}{C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L + C_2 R_2}$
 wo: $\frac{\sqrt{R_2 g_m + 1}}{\sqrt{C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_L}}$
 bandwidth: $\frac{C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L + C_2 R_2}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_2} \sqrt{R_1 + R_L} \sqrt{C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_L}}$

$$\begin{aligned}
&\text{K-LP: } R_L \\
&\text{K-HP: } \frac{R_1 R_L}{R_1 + R_L} \\
&\text{K-BP: } \frac{C_1 R_1 R_2 R_L g_m + C_1 R_1 R_L + C_2 R_2 R_L}{C_1 R_1 R_2 g_m + C_1 R_1 + C_1 R_2 + C_1 R_L + C_2 R_2} \\
&\text{Qz: None} \\
&\text{Wz: } \frac{\sqrt{R_2 g_m + 1}}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2}}
\end{aligned}$$

$$10.4 \quad \mathbf{X-INVALID-WZ-4} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_L g_m + s^2 (C_1 C_2 R_1 R_2 R_L g_m + C_1 C_2 R_1 R_L) + s (C_1 R_1 R_L g_m + C_2 R_2 R_L g_m + C_2 R_L)}{g_m + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_1 C_2 R_2 + C_1 C_2 R_L) + s (C_1 R_1 g_m + C_1 + C_2 R_2 g_m + C_2)}$$

Parameters:

$$\begin{aligned}
&\text{Q: } \frac{\sqrt{C_1} \sqrt{C_2} R_1 R_2 g_m \sqrt{\frac{g_m}{R_1 R_2 g_m + R_1 + R_2 + R_L}} + \sqrt{C_1} \sqrt{C_2} R_1 \sqrt{\frac{g_m}{R_1 R_2 g_m + R_1 + R_2 + R_L}} + \sqrt{C_1} \sqrt{C_2} R_2 \sqrt{\frac{g_m}{R_1 R_2 g_m + R_1 + R_2 + R_L}} + \sqrt{C_1} \sqrt{C_2} R_L \sqrt{\frac{g_m}{R_1 R_2 g_m + R_1 + R_2 + R_L}}}{C_1 R_1 g_m + C_1 + C_2 R_2 g_m + C_2} \\
&\text{wo: } \sqrt{\frac{g_m}{C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_1 C_2 R_2 + C_1 C_2 R_L}} \\
&\text{bandwidth: } \frac{\sqrt{C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_1 C_2 R_2 + C_1 C_2 R_L} (C_1 R_1 g_m + C_1 + C_2 R_2 g_m + C_2)}{\sqrt{C_1} \sqrt{C_2} R_1 R_2 g_m \sqrt{\frac{g_m}{R_1 R_2 g_m + R_1 + R_2 + R_L}} + \sqrt{C_1} \sqrt{C_2} R_1 \sqrt{\frac{g_m}{R_1 R_2 g_m + R_1 + R_2 + R_L}} + \sqrt{C_1} \sqrt{C_2} R_2 \sqrt{\frac{g_m}{R_1 R_2 g_m + R_1 + R_2 + R_L}} + \sqrt{C_1} \sqrt{C_2} R_L \sqrt{\frac{g_m}{R_1 R_2 g_m + R_1 + R_2 + R_L}}} \\
&\text{K-LP: } R_L \\
&\text{K-HP: } \frac{R_1 R_2 R_L g_m + R_1 R_L}{R_1 R_2 g_m + R_1 + R_2 + R_L} \\
&\text{K-BP: } \frac{C_1 R_1 R_L g_m + C_2 R_2 R_L g_m + C_2 R_L}{C_1 R_1 g_m + C_1 + C_2 R_2 g_m + C_2} \\
&\text{Qz: None} \\
&\text{Wz: } \sqrt{\frac{g_m}{C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1}}
\end{aligned}$$

11 X-PolynomialError